Huawei Access Node (KR-ACC-72) SOP for Optical Transport Failure (ALARM_7230)

Node: KR-ACC-72 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_7230

Description

The Optical Transport Failure (ALARM_7230) on Huawei Access Node (KR-ACC-72) is a critical issue that occurs when there is a degradation of SFP/XFP transceivers, fiber cuts, or Rx/Tx power loss in the optical transport domain. This failure domain is specific to the Access role on Huawei equipment due to the inherent design of the optical transport architecture, which relies on the optical signals being transmitted over the fiber-optic cables to facilitate data transmission between nodes. When the optical signals are degraded or lost, the node cannot accurately detect the data being transmitted, resulting in an Optical Transport Failure alarm. The Huawei Access Node (KR-ACC-72) is designed to monitor the optical signals and detect any anomalies, triggering an alarm when an Optical Transport Failure is detected. This SOP outlines the steps to diagnose and resolve the Optical Transport Failure on Huawei Access Node (KR-ACC-72).

Pre-requisite Checks

- Check: Verify that the site is physically accessible and secure, Description: Ensure that the site is not under maintenance or repair, and that all personnel have the necessary clearance and training to access the site.

- Check: Verify that the redundant link is operational, Description: Check that the redundant link is up and functioning correctly to prevent data loss during the diagnostic and resolution process.

- Check: Verify that the Huawei equipment is at the correct firmware version, Description: Ensure that the Huawei equipment is running the latest firmware version to prevent any compatibility issues during the diagnostic and resolution process.

Potential Root Causes

Cause	Probability
Degradation of SFP/XFP transceivers	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display optical link, Expected Output: Link status, Tx power, Rx power, and optical signal-to-noise ratio (SNR), Description: This command displays the link status, Tx power, Rx power, and optical SNR of the optical link, helping to identify any degradation or loss of optical signals.
```

```
Command: display optical transceiver, Expected Output: Transceiver status, temperature, and optical signal quality, Description: This command displays the status of the optical transceiver, temperature, and optical signal quality, helping to identify any issues with the transceiver or its environment.

Command: display optical fiber status, Expected Output: Fiber status, distance, and optical signal quality, Description: This command displays the status of the fiber, distance, and optical signal quality, helping to identify any issues with the fiber or its connections.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP transceiver, Command: replace optical transceiver, Description: Replace the SFP/XFP transceiver with a new one to restore the optical signal transmission. Ensure that the new transceiver is compatible with the Huawei equipment and is properly configured.

Step 2: Step: Repair or replace the fiber, Command: repair optical fiber, Description: Repair or replace the fiber to restore the optical signal transmission. Ensure that the fiber is properly connected and configured to prevent any further issues.

Step 3: Step: Adjust the Rx/Tx power, Command: adjust optical power, Description: Adjust the Rx/Tx power to restore the optical signal transmission. Ensure that the power settings are properly configured to prevent any further issues.

Post-Check Verification

```
Command: display optical link, Expected Output: Link status, Tx power, Rx power, and optical SNR, Description: Run this command to verify that the optical link is stable and that the Tx power, Rx power, and optical SNR are within the normal range.

Command: display optical transceiver, Expected Output: Transceiver status, temperature, and optical signal quality, Description: Run this command to verify that the optical transceiver is stable and that the temperature and optical signal quality are within the normal range.

Command: display optical fiber status, Expected Output: Fiber status, distance, and optical signal quality, Description: Run this command to verify that the fiber is stable and that the distance and optical signal quality are within the normal range.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP transceiver, Command: restore original optical transceiver, Description: Restore the original SFP/XFP transceiver to its previous state to prevent any further issues.

- Step: Restore the original fiber configuration, Command: restore original optical fiber configuration, Description: Restore the original fiber configuration to its previous state to prevent any further issues.

- Step: Restore the original Rx/Tx power settings, Command: restore original optical power settings, Description: Restore the original Rx/Tx power settings to their previous state to prevent any further issues.